

# Macrosopes for Making Sense of Science and Technology

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## Summary

The *Places & Spaces: Mapping Science* exhibit introduces visualizations of the evolving science and technology (S&T) ‘landscape’ to a general audience. The maps show the structure and interconnections between scientific disciplines, the birth of new ‘lands’ of science or technology, and the diffusion of ideas across the landscape of S&T. Each iteration showcases the benefits of data visualization for a particular audience, e.g., for economic decision makers, science policy makers, scholars, librarians, or kids. At its heart, the exhibit’s goal is to promote validated and replicable workflows for the design of data visualization and to increase public understanding of the power of S&T maps to help us make sense of the increasingly large datasets that we all face on a daily basis.

The exhibition includes 100 science maps plus sculptures, hands-on activities, and interactive visualizations called macrosopes. Each year, four new macrosopes are chosen to travel with the exhibit. These macrosopes are accessible to museum visitors via a touch screen kiosk [Figure 1].

These four new macrosopes use data that varies tremendously in terms of subject matter and method of data collection: city smells geotagged in social media posts [1], a digital library collection visualized using a flowmap to connect places of publication with potential readers [2], networks of institutional partnerships revealed through publications [3], and marine traffic tracked via satellite and visualized at global and local scales [4]. Collectively, the macrosopes showcase effective means to visually communicate massive datasets to gain new perspectives and to ask new questions. Macroscope users are invited to touch and manipulate the data visualizations to gain a more comprehensive understanding of complex socio-technical systems.

Drawing from across cultures and across scholarly disciplines, the *Places & Spaces: Mapping Science* exhibit demonstrates the power of maps to address vital questions about the contours and content of human knowledge. Created by leading experts in the natural, physical, social behavioral and economic sciences as well as the visual arts and humanities, works represented in *Places & Spaces* provide a better grasp of the abstract contexts, relationships, and dynamism of natural and human-constructed systems and collective intelligence.

## References

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**Figure 1:** Visitors at the David J. Sencer CDC Museum explore the macroscopes.  
Photograph courtesy of Mike Jensen.